

Banking Transaction Management System (BTMS)

Database Testing

TEST PLAN

1. Project Overview

BTMS manages customer accounts, financial transactions (deposits, withdrawals, transfers, bill payments), transaction categories, fees, and branch/employee records. Testing targets 7 core tables: Customers, Accounts, Transaction_Categories, Transactions, Fees, Branches, and Employees, built on MySQL 8.0.

2. Objectives

- Verify that all mandatory fields, uniqueness rules, and data types are enforced correctly.
- Verify referential integrity across all foreign-key relationships.
- Verify CHECK constraints on numeric fields (balance, amount, fees).
- Verify business rules implemented via triggers (balance updates, fee calculation, daily transfer limits).
- Identify and report defects with severity classification to support prioritized fixes.

3. Scope

3.1 In-Scope

- Customer accounts (creation, update, uniqueness of email)
- Financial transactions (deposits, withdrawals, transfers, bill payments)
- Transaction categories
- Branches and employees
- Fees and transaction limits
- Account balance management (automatic updates via triggers)

3.2 Out-of-Scope

- Front-end UI
- APIs
- Reporting dashboards

3.3 Testing Scope

- INSERT and UPDATE queries only
- SQL validation of business rules
- Data integrity checks

4. Test Types

- Functional Testing — validating that each functional requirement (FR-1 to FR-6) behaves as specified.
- Database Testing — schema validation, constraints, triggers, referential integrity.
- Positive Testing — valid data accepted and processed correctly.
- Negative Testing — invalid/edge-case data correctly rejected (negative amounts, duplicate emails, overdrafts, invalid enum values, transfers exceeding daily limits, missing receiver accounts).

5. Test Environment

Item	Detail
Operating System	Windows 11
Database	MySQL 8.0.43
Test Data	Synthetic sample data for Customers, Accounts, Transactions, Categories, Fees, Branches, Employees
Tools	MySQL Workbench / SQL client, Jira (BTMS project) for test case and defect tracking

6. Entry Criteria

- Database schema (7 tables) created and deployed to the test environment.
- Triggers for balance updates and fee calculation implemented.
- BRD and functional requirements (FR-1 to FR-6) reviewed and approved.
- Test cases written and reviewed.

7. Exit Criteria

- All planned test cases (119) executed.
- Pass rate of at least 95% on executed test cases..
- No open Critical/High severity defects